

14/04/2025

CODE-A



# Aakash

Medical | IIT-JEE | Foundations

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Sector-18, Udyog Vihar, Gurugram, Haryana-122015



## AIM - 720

### (Advanced INTENSIVE Mastery for 720)

MM : 720

CST-4

Time : 180 Mins.

**Complete Syllabus of NEET**

#### Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Physics, Chemistry, Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

#### 6 Aakashians scored a perfect 720/720 from Classroom Programs



720/720 **Mridul M Anand**  
3 Year Classroom



720/720 **Ayush Naugraiya**  
4 Year Classroom



720/720 **Arghyadeep Dutta**  
2 Year Classroom



720/720 **Aryan Yadav**  
1 Year Classroom



720/720 **Palansha Agarwal**  
2 Year Classroom



720/720 **Iram Quazi**  
1 Year Classroom

PHYSICS

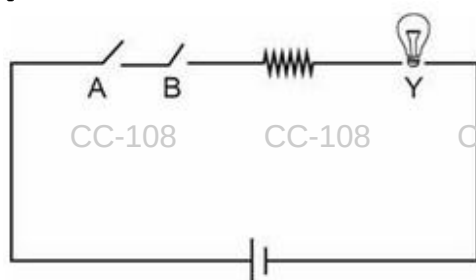
1. A nucleus of nuclear density  $\rho$  disintegrates into two daughter nuclei with masses in the ratio 9:21. Density of the smaller nucleus is

- (1)  $\frac{2}{3}\rho$
- (2)  $\frac{3}{9}\rho$
- (3)  $\frac{9}{21}\rho$
- (4)  $\rho$

2. The ionisation potential of hydrogen is 13.6 V. The energy required to remove an electron from the third orbit of hydrogen is

- (1) 3.4 eV
- (2) 1.51 eV
- (3) 12.09 eV
- (4) 12.75 eV

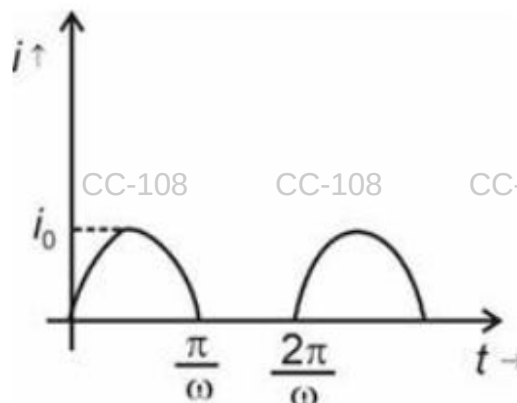
3. The circuit diagram shown here corresponds to the logic gate



- (1) AND gate
  - (2) NAND gate
  - (3) NOR gate
  - (4) OR gate
4. The amplitude of oscillations is reduced to half of its initial value of 40 cm due to damping by resistive force in time  $t_0$ , then after time  $3t_0$ , its amplitude will be

- (1) 5 cm
- (2) 10 cm
- (3) 2.5 cm
- (4)  $\frac{20}{3}$  cm

5. The output current versus time curve of a half wave rectifier is shown in the figure. The average value of the output current from  $t = 0$  to  $t = \frac{\pi}{\omega}$  is (where input signal is given by,  $i = i_0 \sin \omega t$ )

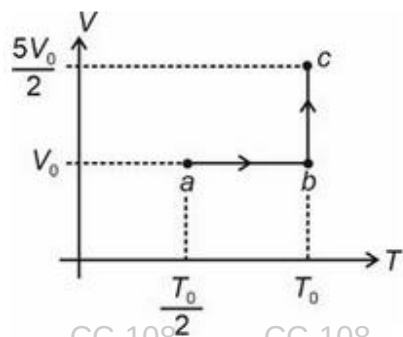


- (1) Zero
  - (2)  $\frac{i_0}{\pi}$
  - (3)  $\frac{2i_0}{\pi}$
  - (4)  $i_0$
6. Given below are two statements  
**Statement-I** : The de Broglie wavelength ( $\lambda$ ) associated with a moving particle is related to its momentum ( $p$ ) as  $\lambda = \frac{h}{p}$   
**Statement-II** : Photons are electrically neutral and are not deflected by electric and magnetic fields.  
 In the light of above statements choose the most appropriate answer from the options given below.
- (1) Statement-I is correct but statement-II is incorrect
  - (2) Statement-I is incorrect but statement-II is correct
  - (3) Both statements-I and II are correct
  - (4) Both statements-I and II are incorrect

7. Two rods one made of material A and other made of material B of same length and same cross-sectional area are joined together. If thermal conductivity of material A is  $K_1$  while that of material B is  $K_2$  and the free end of rod made of material A is maintained at  $T_1$  while that of the rod of material B is maintained at  $T_2$ , then the temperature of junction is (Where  $T_1 > T_2$ )

- (1)  $\frac{T_1+T_2}{2}$
- (2)  $\frac{T_1K_1+T_2K_2}{K_1+K_2}$
- (3)  $\frac{T_1K_1-T_2K_2}{K_1+K_2}$
- (4)  $\frac{T_1K_1+T_2K_2}{K_1-K_2}$

8. One mole of an ideal monoatomic gas is taken through process  $a \rightarrow b \rightarrow c$  as shown in  $V-T$  curve. (Where  $V$  represents volume and  $T$  represents temperature).



The change in internal energy in process  $a \rightarrow b$  will be

- (1) Zero
- (2)  $\frac{3RT_0}{4}$
- (3)  $\frac{5RT_0}{4}$
- (4)  $\frac{4RT_0}{3}$

9. Consider the following statements out of which one is labelled as assertion and other as reason.

**Assertion :** The internal energy of an ideal monoatomic gas enclosed in a container does not change when there is no change in temperature.

**Reason :** Internal energy of a gaseous system is path function.

- (1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true and Reason (R) is false.
- (4) Assertion (A) is false and Reason (R) is true.

10. A flask contains hydrogen and oxygen gas in the ratio of 3 : 1 by mass at temperature  $27^\circ\text{C}$ . The ratio of average translational kinetic energy per molecule of hydrogen and oxygen respectively is

- (1) 1 : 1
- (2) 3 : 1
- (3) 1 : 3
- (4) 1 : 4

11. Two copper vessels A and B have the same base area but of different shapes. Vessel A takes twice the volume of water as that vessel B requires to fill upto common particular height. Then correct statement among the following is

- (1) Force due to liquid on base of vessel A is more
- (2) Force due to liquid on base of vessel B is more
- (3) Pressure at base of vessel A is more
- (4) Pressure at base of both the vessels is same

12. A small iron needle placed slowly on the surface of water floats due to

- (1) Surface tension
- (2) Viscosity
- (3) Gravity
- (4) Both (2) & (3)

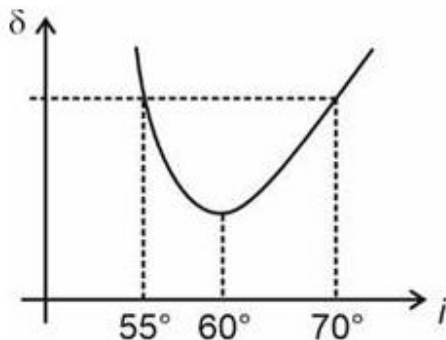
13. A particle is moving in x-y plane such that its x and y coordinates changes with time according to relation,  $x = 3t^2$  &  $y = 5t$  (here x & y are in m & t is in s). Speed of the particle at  $t = 2$  s, will be

- (1)  $17 \text{ m s}^{-1}$
- (2)  $\sqrt{34} \text{ m s}^{-1}$
- (3)  $13 \text{ m s}^{-1}$
- (4)  $11 \text{ m s}^{-1}$

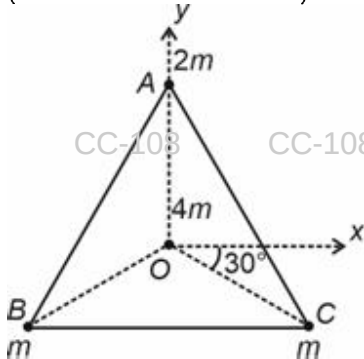
14. A rigid body rotates about a stationary axis according to the equation,  $\theta = 6t^2 - 2t^3$ . Its angular velocity, when its angular acceleration becomes  $6 \text{ rad/s}^2$ , will be (Where  $\theta$  denotes angular displacement in radian, t denotes time in second)

- (1) Zero
- (2)  $3 \text{ rad/s}$
- (3)  $4.5 \text{ rad/s}$
- (4)  $6 \text{ rad/s}$

15. A graph is plotted between angle of deviation ( $\delta$ ) in an equilateral prism and angle of incidence ( $i$ ) as shown in the figure. Refractive index of the prism is



- (1)  $\sqrt{2}$   
 (2) 3.5  
 (3)  $\sqrt{3}$   
 (4) 2
16. Three masses  $2m$ ,  $m$  and  $m$  are fixed at the vertices of an equilateral triangle  $ABC$  as shown below. The force acting on a mass  $4m$  placed at the centroid  $O$  of the triangle, is (Given  $AO = BO = CO = 2$  m)



- (1)  $3Gm^2\hat{j}$   
 (2)  $Gm^2\hat{j}$   
 (3)  $-3Gm^2\hat{j}$   
 (4)  $-Gm^2\hat{j}$
17. Given below are two statements :  
**Statement-I** : If a parallel paraxial beam of light were incident on a concave mirror, making some angle with the principal axis, the reflected rays would converge at a point in focal plane.  
**Statement-II** : Power( $P$ ) of concave mirror is positive even though focal length( $f$ ) is negative.  
 In the light of the above statements, choose the most appropriate answer from the options given below.
- (1) Both statement I and statement II are correct  
 (2) Both statement I and statement II are incorrect  
 (3) Statement I is correct but statement II is incorrect  
 (4) Statement I is incorrect but statement II is correct

18. Keeping the mass of earth to be constant if its radius changes from  $R$  to  $\frac{R}{\sqrt{3}}$ , then new duration of the day will be

- (1) 6 hrs  
 (2) 8 hrs  
 (3)  $8\sqrt{3}$  hrs  
 (4)  $24\sqrt{3}$  hrs

19. Which of the following statements are true about acceleration due to gravity ( $g$ )? (where symbols have their usual meaning)?

- (1)  $g$  will not change at poles if earth stops rotating on its axis  
 (2) The value of acceleration due to gravity decreases as we move away from surface of earth  
 (3) The value of acceleration due to gravity decreases as we move from surface towards centre of earth  
 (4) All of these

20. The Young's modulus of wire of length  $L$  and radius  $r$  is  $Y$ . If the length and radius are reduced to  $\frac{L}{3}$  and  $\frac{r}{2}$ , then its Young's modulus will be

- (1)  $\frac{3}{4}Y$   
 (2)  $Y$   
 (3)  $6Y$

- (4)  $\frac{2}{3}Y$

21. A block is resting on a piston which is moving vertically executing SHM of period 1 s. At what minimum amplitude of motion, will the block and piston separate? (take  $\pi^2 = 10$ )

- (1) 0.2 m  
 (2) 0.3 m  
 (3) 0.25 m  
 (4) 0.5 m

22. Which of the following equations represents a progressive harmonic wave travelling along y-axis? (All symbols have their usual meaning)

- (1)  $y = A \sin(kz + \omega t)$   
 (2)  $y = A \sin kx \cos 2\pi t$   
 (3)  $x = A \sin(ky - \omega t)$   
 (4)  $x = A \sin(kz + \omega t)$

23. The velocity of an electromagnetic wave is along (where  $E$  represents the electric field and  $B$  represents the magnetic field)

- (1)  $\vec{E} + \vec{B}$
- (2)  $\vec{E} \times \vec{B}$
- (3)  $\vec{E}$
- (4)  $\vec{B}$

24. The Brewster's angle  $\theta$  for an air-medium interface should be

- (1)  $0^\circ < \theta < 30^\circ$
- (2)  $30^\circ < \theta < 45^\circ$
- (3)  $45^\circ < \theta < 90^\circ$
- (4)  $\theta = 90^\circ$

25. According to Gauss's law in electrostatics, net electric flux through a closed surface depends on

- (1) Shape of the surface
- (2) Area of the surface
- (3) Quantity of charge enclosed by the surface
- (4) All of these

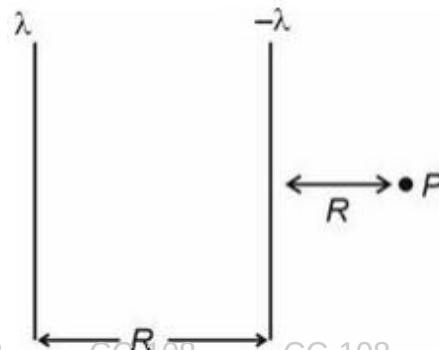
26. An electric dipole is placed in a uniform electric field making an angle  $30^\circ$  with electric field and it experiences torque equal to  $\tau$  in this scenario. The minimum work done in changing the orientation from  $30^\circ$  to  $60^\circ$  is equal to

- (1)  $(\sqrt{3}-1)\tau$
- (2)  $\frac{(\sqrt{3}-1)\tau}{2}$
- (3)  $(\sqrt{2}+1)\tau$
- (4)  $\frac{(\sqrt{2}+1)\tau}{2}$

27. A copper wire of length 1 m and radius  $\left(\frac{10^{-3}}{\sqrt{\pi}}\right)$  m has electrical resistance of  $10 \Omega$ . The current density in the wire for an electric field strength of  $10 \text{ V/m}$  is

- (1)  $10^5 \text{ A/m}^2$
- (2)  $10^6 \text{ A/m}^2$
- (3)  $2 \times 10^5 \text{ A/m}^2$
- (4)  $2 \times 10^6 \text{ A/m}^2$

28. Two parallel infinite line charge having linear charge densities as  $\lambda$  and  $-\lambda$  are placed at a distance of  $R$  in free space as shown. The magnitude of net electric field at point  $P$  is



- (1)  $\frac{\lambda}{2\pi\epsilon_0 R}$
- (2)  $\frac{\lambda}{4\pi\epsilon_0 R}$
- (3)  $\frac{2\lambda}{\pi\epsilon_0 R}$
- (4)  $\frac{3\lambda}{2\pi\epsilon_0 R}$

29. If electric potential (in volt) in a region is expressed as  $V(x, y, z) = 2xy - yz$ , the electric field (in N/C) at point  $(1, 0, 1)$  will be equal to

- (1)  $-\hat{j}$
- (2)  $2\hat{i} - \hat{j}$
- (3)  $-\hat{j} + \hat{k}$
- (4)  $\hat{i} - \hat{j} + \hat{k}$

30. Consider the following statements and choose the correct option.

**Statement (A)** : Kirchhoff's junction law follows the conservation of charge.

**Statement (B)** : Kirchhoff's loop law follows the conservation of energy.

- (1) Only statement (A) is correct
- (2) Only statement (B) is correct
- (3) Both statements are correct
- (4) Both statements are incorrect

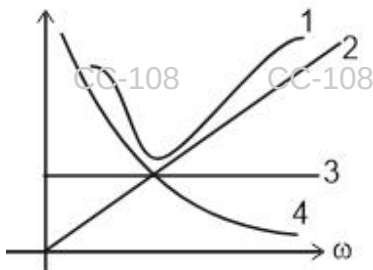
31. Induced emf produced in a coil of area  $A$  rotating with angular velocity  $\omega$  in a magnetic field  $B$  is equal to  $\frac{\sqrt{3}BA\omega}{2}$ , when the angle between the area vector of the coil and direction of magnetic field is

- (1)  $\frac{\pi}{3}$
- (2)  $\frac{\pi}{6}$
- (3)  $\pi$
- (4) Zero

32. An a.c. voltage given by relation,  $V = 300\sin(100t)$  volt is connected with resistor having resistance  $60\ \Omega$  and inductor of inductance  $L$ . If peak current in the circuit is 3 A, the value of  $L$  will be (where  $t$  denotes the time in s)

(1) 0.6 H  
(2) 0.4 H  
(3) 0.8 H  
(4) 1 H

33. The variation of different physical quantities in a series LCR circuit with angular frequency ( $\omega$ ) of AC source is shown below.

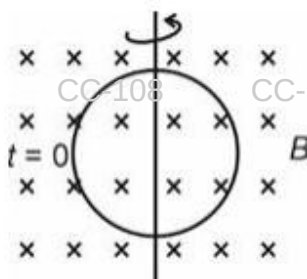


Based on above information, match the two columns.

| Column-I                 | Column-II |
|--------------------------|-----------|
| (a) Resistance           | (p) 4     |
| (b) Capacitive reactance | (q) 1     |
| (c) Inductive reactance  | (r) 2     |
| (d) Impedance            | (s) 3     |

(1)  $a \rightarrow s, b \rightarrow p, c \rightarrow r, d \rightarrow q$   
(2)  $a \rightarrow q, b \rightarrow p, c \rightarrow r, d \rightarrow s$   
(3)  $a \rightarrow q, b \rightarrow r, c \rightarrow p, d \rightarrow s$   
(4)  $a \rightarrow s, b \rightarrow p, c \rightarrow q, d \rightarrow r$

34. A circular loop of radius  $R$  is rotating about its diameter with angular velocity  $\omega$  in uniform magnetic field  $B$  as shown. If at  $t = 0$ , plane of the loop is perpendicular to the magnetic field, then the average e.m.f induced in time interval  $(\frac{T}{2})$  will be (where  $T$  denotes the time period)



(1)  $R^2\omega B$   
(2)  $2R^2B\omega$   
(3)  $\frac{R^2\omega B}{2}$   
(4) Zero

35. Two long thin parallel wires separated by a distance  $d$  are carrying a current  $i$  each. The magnitude of the force per unit length applied by one wire on the other is

(1) Directly proportional to  $(i)^2$   
(2) Inversely proportional to  $(i)^2$   
(3) Directly proportional to  $i$   
(4) Inversely proportional to  $i$

36. A charged particle is moving on circular path with velocity  $v$  in a uniform magnetic field  $B$ . If velocity of the particle and strength of magnetic field is doubled, then time taken to complete one revolution becomes

(1) 8 times  
(2) 4 times  
(3)  $\frac{1}{2}$  times  
(4)  $\frac{1}{8}$  times

37. A material suitable for making electromagnets should have

(1) High retentivity and low coercivity  
(2) High retentivity and high coercivity  
(3) Low retentivity and low coercivity  
(4) Low retentivity and high coercivity

38. Choose the quantities that are dimensionless among the following.

(1)  $\frac{\text{Torque}}{\text{Force}}$

(2)  $\frac{\text{Torque}}{\text{Work}}$

(3)  $\frac{\text{Work}}{\text{Energy}}$

(4) Both (2) and (3)

39. Given below are two statements one is labelled as assertion (A) and Reason (R)

**Assertion:** The length of actual path travelled by a body in given time interval is always equal to displacement.

**Reason:** If displacement is zero, then body is either at rest or it has returned to initial position. Choose the correct option.

(1) Assertion is true but Reason is false  
(2) Assertion is false but Reason is true  
(3) Both Assertion and Reason are true and Reason is correct explanation of Assertion  
(4) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion

40. A truck moving with velocity 36 km/hr is stopped by applying brakes in 2 s. If same truck moves with speed 144 km/h and brakes are applied then the stopping time will be (Assume the same retardation in both cases)

(1) 2 s  
(2) 5 s  
(3) 9 s  
(4) 8 s

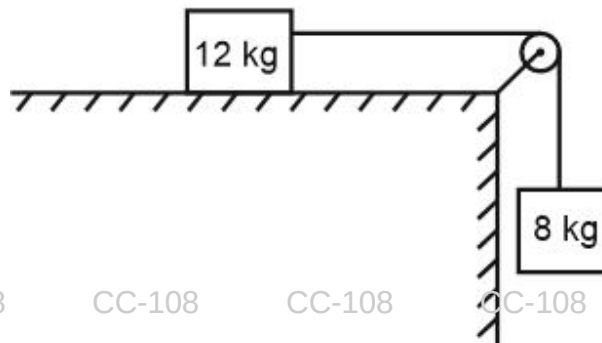
41. Two physical quantities  $P$  and  $Q$  have different dimensions. The mathematical operation that is not possible among the following are

(1)  $\frac{P}{Q}$   
(2)  $PQ$   
(3)  $P + Q$   
(4) Both (1) and (2)

42. A spring of length  $L$  and spring constant  $K$  is cut into two parts of length  $\frac{L}{3}$  and  $\frac{2L}{3}$ , then the spring constant of each part will be

(1)  $3K, \frac{3K}{2}$   
(2)  $2K, \frac{2K}{3}$   
(3)  $K, \frac{K}{3}$   
(4)  $K, K$

43. A block of mass 12 kg is kept on a rough horizontal surface. It is connected to a string, which passes over a pulley. When a weight of 8 kg is attached to the other end of the string, then the block just begins to move, then coefficient of static friction is equal to



(1)  $\frac{3}{2}$   
(2)  $\frac{2}{3}$   
(3)  $\frac{1}{2}$   
(4)  $\frac{1}{3}$

44. A particle of mass 0.4 kg is moving with a velocity of  $(6\hat{i} - 8\hat{j})$  m/s, then the kinetic energy of the particle is

(1) 10 J  
(2) 20 J  
(3) 30 J  
(4) Zero

45. A force  $F = (-0.6x^2 + 2.5)$  N displaces a particle of mass 2 kg from  $x = 0$  to  $x = 10$  m. If velocity of particle at  $x = 0$  was 20 m/s, then velocity at  $x = 10$  m will be

(1) 10 m/s  
(2) 20 m/s  
(3) 15 m/s  
(4) 60 m/s

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CHEMISTRY

46. Ethylenediamine tetraacetate ion is a/an:

(1) Monodentate ligand  
(2) Bidentate ligand  
(3) Ambidentate ligand  
(4) Hexadentate ligand

47. Given below are two statements

**Statement I:**  $[\text{Ni}(\text{CO})_4]$  has longer C – O bond length than  $[\text{Co}(\text{CO})_4]^-$ .

**Statement II:**  $[\text{Ni}(\text{CO})_4]$  has stronger metal carbon (M – C) bond than  $[\text{Co}(\text{CO})_4]^-$ .

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and Statement II are correct
- (2) Both statement I and Statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

48. Given below are two statements: one is labelled as Assertion (A) and other is labelled as reason (R).

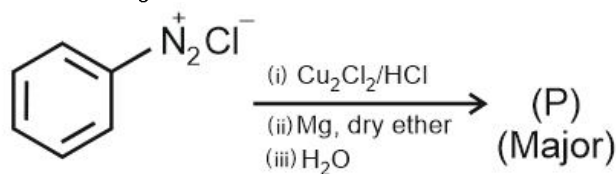
**Assertion (A) :** Haloarenes are less reactive towards nucleophilic substitution reaction than haloalkanes.

**Reason (R) :** In haloarenes, C – X (X = halogen) bond acquires a partial double bond character due to resonance, as a result the bond cleavage in haloarenes is difficult than haloalkanes.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

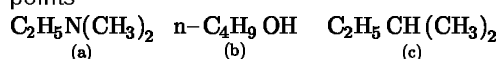
49. In the following reaction



The product (P) will be

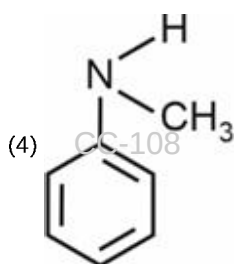
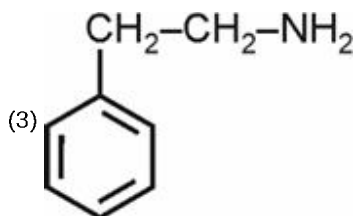
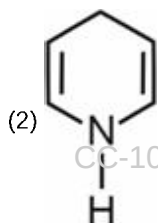
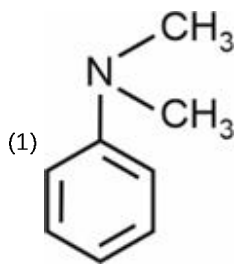
- (1)
- (2)
- (3)
- (4)

50. Arrange the following in their decreasing order of boiling points



- (1) (b) > (a) > (c)
- (2) (a) > (b) > (c)
- (3) (c) > (a) > (b)
- (4) (b) > (c) > (a)

51. Which of the following amines gives carbylamine reaction?



52. Match List-I with List-II

| List I<br>(Battery)                      | List-II<br>(Electrolyte)                                       |
|------------------------------------------|----------------------------------------------------------------|
| a. Dry cell                              | (i) Concentrated aqueous NaOH solution                         |
| b. Lead storage battery                  | (ii) Moist paste of $\text{NH}_4\text{Cl}$ and $\text{ZnCl}_2$ |
| c. Mercury cell                          | (iii) 38% solution of $\text{H}_2\text{SO}_4$                  |
| d. $\text{H}_2$ - $\text{O}_2$ fuel cell | (iv) Paste of $\text{KOH}$ & $\text{ZnO}$                      |

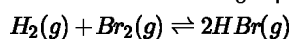
The correct match is

- (1) a(ii), b(i), c(iii), d(iv)  
 (2) a(iv), b(iii), c(i), d(ii)  
 (3) a(ii), b(iii), c(iv), d(i)  
 (4) a(i), b(ii), c(iv), d(iii)

53. Calculate the ratio of concentration of sodium acetate to acetic acid which is required to get a buffer of pH 6.76 ( $\text{pK}_a$  of  $\text{CH}_3\text{COOH} = 4.76$ ).

- (1) 1 : 10  
 (2) 1 : 100  
 (3) 100 : 1  
 (4) 10 : 1

54. Consider the following equilibrium



If  $K_C$  of the reaction is  $1.6 \times 10^5$  at 1024 K, then  $K_P$  of the reaction will be ( $R = 0.0821 \text{ atm L K}^{-1} \text{ mol}^{-1}$ )

- (1)  $6.7 \times 10^6$   
 (2)  $1.34 \times 10^7$   
 (3)  $1.6 \times 10^5$   
 (4)  $1.34 \times 10^5$

55. Electrolysis of molten calcium sulphate solution was carried out by passing 9.65 amperes current. The time required to deposit 0.2 g of calcium at the cathode is

- (1) 88 s  
 (2) 10 s  
 (3) 90 s  
 (4) 100 s

56. Eka – Aluminium and Eka-silicon respectively are

- (1) Beryllium and calcium  
 (2) Boron and scandium  
 (3) Gallium and germanium  
 (4) Germanium and gallium

57. Given below are two statements

**Statement I:** The elements after uranium are called Transuranium elements.

**Statement II:** The IUPAC nomenclature for atomic number 112 is Ununbium.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct  
 (2) Both statement I and Statement II are incorrect  
 (3) Statement I is correct but statement II is incorrect  
 (4) Statement I is incorrect but statement II is correct

58. The correct order of ionic character is

- (1)  $\text{KCl} < \text{CaCl}_2 < \text{ZnCl}_2$   
 (2)  $\text{ZnCl}_2 < \text{CaCl}_2 < \text{KCl}$   
 (3)  $\text{KCl} < \text{ZnCl}_2 < \text{CaCl}_2$   
 (4)  $\text{ZnCl}_2 < \text{KCl} < \text{CaCl}_2$

59. The correct order of bond length among the following is

- (1)  $C-N < C=N < C \equiv N$
- (2)  $C=N < C \equiv N < C-N$
- (3)  $C \equiv N < C=N < C-N$
- (4)  $C-N < C \equiv N < C=N$

60. Match species given in List-I with their bond angles in List-II

|    | List-I |       | List-II       |
|----|--------|-------|---------------|
| a. | $CH_4$ | (i)   | $107^\circ$   |
| b. | $NH_3$ | (ii)  | $109.5^\circ$ |
| c. | $BF_3$ | (iii) | $104.5^\circ$ |
| d. | $H_2O$ | (iv)  | $120^\circ$   |

Choose the correct option

- (1) a(ii), b(i), c(iv), d(iii)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(ii), b(iv), c(i), d(iii)
- (4) a(iv), b(ii), c(iii), d(i)

61. The vapour pressure of pure liquids A and B are 400 and 600 mmHg, respectively at 300 K. Calculate the mole fraction of A and B in liquid mixture if total vapour pressure is 450 mmHg.

- (1)  $X_A = 0.75, X_B = 0.25$
- (2)  $X_A = 0.25, X_B = 0.75$
- (3)  $X_A = 0.60, X_B = 0.40$
- (4)  $X_A = 0.40, X_B = 0.60$

62. A solution of NaCl is isotonic with 0.2 M urea solution. The molarity of NaCl solution will be

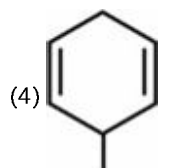
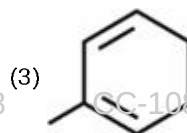
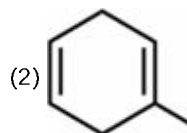
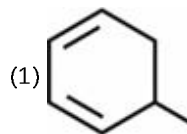
- (1) 0.4 M
- (2) 0.3 M
- (3) 0.2 M
- (4) 0.1 M

63. Correct set of four quantum numbers for the valence electron of potassium atom is

|     | n | l | $m_l$ | $m_s$          |
|-----|---|---|-------|----------------|
| (1) | 4 | 1 | 1     | $+\frac{1}{2}$ |
| (2) | 4 | 1 | 0     | $+\frac{1}{2}$ |
| (3) | 3 | 0 | 0     | $+\frac{1}{2}$ |
| (4) | 4 | 0 | 0     | $+\frac{1}{2}$ |

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

64. Which compound on reductive ozonolysis gives methyl glyoxal as one of the products?



65. Match List-I with List-II

|    | List-I  |       | List-II |
|----|---------|-------|---------|
| a. | Thymine | (i)   |         |
| b. | Uracil  | (ii)  |         |
| c. | Guanine | (iii) |         |
| d. | Adenine | (iv)  |         |

Choose the correct answer from the options given below

- (1) a(ii), b(iii), c(iv), d(i)
- (2) a(iii), b(ii), c(iv), d(i)
- (3) a(ii), b(iii), c(i), d(iv)
- (4) a(iv), b(iii), c(ii), d(i)

66. Correct order of energy for 2s orbital for the given elements is
- (1)  $K > Na > Li > H$
  - (2)  $H > K > Na > Li$
  - (3)  $Li > H > Na > K$
  - (4)  $H > Li > Na > K$
67. Consider the following statements
- (a) Xerophthalmia is caused by the deficiency of vitamin A
  - (b) B group vitamins and vitamin C are soluble in water
  - (c) Vitamin K is a fat-soluble vitamin
- The correct statements are
- (1) (a) and (b) only
  - (2) (b) and (c) only
  - (3) (a) and (c) only
  - (4) (a), (b) and (c)
68. Given below are the two statements
- Statement I:** Melting point of cis-But-2-ene is higher than trans-But-2-ene.
- Statement II:** Dipole moment of cis-But-2-ene is higher than trans-But-2-ene
- In the light of above statements, choose the correct answer
- (1) Statement I is correct but statement II is incorrect
  - (2) Statement I is incorrect but statement II is correct
  - (3) Both statement I and statement II are correct
  - (4) Both statement I and Statement II are incorrect
69. Haemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 5.6 g of haemoglobin is
- (1)  $3.4 \times 10^{18}$
  - (2)  $2.04 \times 10^{20}$
  - (3)  $3.4 \times 10^{16}$
  - (4)  $2.04 \times 10^{18}$
70. Spin only magnetic moment is same for which of the following ions?
- A.  $Ti^{3+}$
  - B.  $Sc^{3+}$
  - C.  $Mn^{4+}$
  - D.  $Fe^{3+}$
  - E.  $Cr^{3+}$
- Choose the most appropriate answer from the options given below:
- (1) D and E only
  - (2) B and C only
  - (3) A and E only
  - (4) C and E only
71. Given below are the two statements
- Statement I:** Permanganate ion is purple in colour whereas manganate ion is green
- Statement II:**  $KMnO_4$  decomposes at 513 K to give  $K_2MnO_4$  and  $MnO_2$  only.
- In the light of above statements, choose the correct answer
- (1) Statement I is correct but statement II is incorrect
  - (2) Statement I is incorrect but statement II is correct
  - (3) Both statement I and statement II are correct
  - (4) Both statement I and Statement II are incorrect
72. For which of the following cases, change in entropy is negative?
- (1) Boiling of an egg
  - (2) Expansion of a gas at constant temperature
  - (3) Stretching of a rubber
  - (4) Evaporation of water
73. At  $25^\circ C$  and 1 atm pressure, for the reaction:  $3C_2H_2(g) \rightarrow C_6H_6(l)$  the enthalpy of reactions is  $-632 \text{ kJ mol}^{-1}$ . The enthalpy of combustion of benzene (l) is  $-3268 \text{ kJ mol}^{-1}$ . The enthalpy of combustion for acetylene (g) is
- (1)  $+1300 \text{ kJ mol}^{-1}$
  - (2)  $-1300 \text{ kJ mol}^{-1}$
  - (3)  $-1500 \text{ kJ mol}^{-1}$
  - (4)  $+1500 \text{ kJ mol}^{-1}$
74. The oxidation states of three bromine atoms present in  $Br_3O_8$  are
- (1) +6, 0, +6
  - (2) -4, 0, +4
  - (3) +6, +4, +6
  - (4) +4, 0, +6
75. Consider the following statements:
- Statement I:** Thermally  $GeCl_4$  is more stable than  $GeCl_2$ .
- Statement II:** Thermally  $PbCl_2$  is more stable than  $PbCl_4$ .
- In the light of above statements, choose the correct option.
- (1) Both statement I and Statement II are correct
  - (2) Both statement I and Statement II are incorrect
  - (3) Statement I is correct but statement II is incorrect
  - (4) Statement I is incorrect but statement II is correct
76. All of the following are heterocyclic aromatic compounds, except
- (1) Furan
  - (2) Thiophene
  - (3) Pyridine
  - (4) Tropone

77. In Dumas method of estimation of nitrogen 0.28 g of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm pressure. The percentage composition of nitrogen in the compound would be (Aqueous tension at 300 K = 15 mm).

- (1) 33.6%
- (2) 11.2%
- (3) 44.8%
- (4) 22.4%

78. Consider the following statements:

- (i) Tertbutyl carbocation is more stable as compared to isopropyl carbocation.
- (ii) Actual structure of nitromethane is a resonance hybrid of three canonical forms.
- (iii) Resonance effect and hyperconjugation both are permanent effects.

Choose the incorrect statement(s).

- (1) (i) and (ii) only
- (2) (ii) and (iii) only
- (3) (i) only
- (4) (ii) only

79. Which of the following contain N – N bond?

- (a)  $N_2O_3$
- (b)  $N_2O_5$
- (c)  $N_2O_4$
- (d)  $N_2O$

Select the correct option

- (1) (a), (b) and (c) only
- (2) (b), (c) and (d) only
- (3) (a), (c) and (d) only
- (4) (a), (b), (c) and (d)

80. Which of the following statement is incorrect?

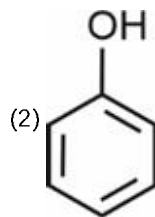
- (1) White phosphorus is insoluble in water but soluble in carbon disulphide.
- (2) White phosphorus readily catches fire in air to give dense white fumes of  $P_4O_{10}$ .
- (3) The solution of phosphine in water decomposes in presence of light giving red phosphorus and  $H_2$ .
- (4) Chemically, white phosphorus is much less reactive than red phosphorus.

81. Which of the following oxoacids of iodine is least likely to exist?

- (1)  $HOIO_2$
- (2)  $HOIO_3$
- (3)  $HOIO$
- (4)  $HOI$

82. The alcohol which react fastest with Lucas reagent is

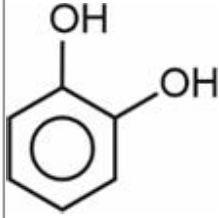
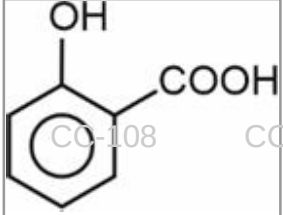
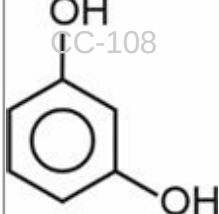
- (1)  $CH_3OH$



- (3)  $(CH_3)_2CH-OH$

- (4)  $(CH_3)_3C-OH$

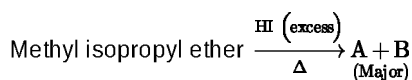
83. Match List-I with List-II

|    | List-I<br>(Structure)                                                                |       | List-II<br>(Common name) |
|----|--------------------------------------------------------------------------------------|-------|--------------------------|
| a. |    | (i)   | Resorcinol               |
| b. |   | (ii)  | Quinol                   |
| c. |  | (iii) | Catechol                 |
| d. |  | (iv)  | Salicylic acid           |

Choose the correct answer from the options given below

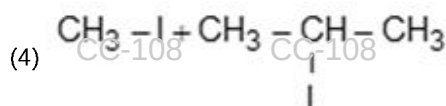
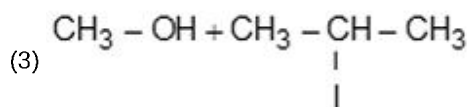
- (1) a(iv), b(iii), c(i), d(ii)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(iii), b(iv), c(ii), d(i)
- (4) a(iii), b(iv), c(i), d(ii)

84. Consider the following reaction



Products A and B are

- (1)  $\text{CH}_3 - \text{I} + \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{I}$   
 (2)  $\text{CH}_3 - \text{I} + \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$



85. Match time in List-I with their values in List-II for first order reaction.

|    | List I       |       | List-II              |
|----|--------------|-------|----------------------|
| a. | $t_{1/2}$    | (i)   | $\frac{\ln 4}{k}$    |
| b. | $t_{75\%}$   | (ii)  | $\frac{\ln 2}{k}$    |
| c. | $t_{99.9\%}$ | (iii) | $\frac{\ln 10}{k}$   |
| d. | $t_{90\%}$   | (iv)  | $\frac{10 \ln 2}{k}$ |

Correct match is

- (1) a(ii), b(i), c(iii), d(iv)  
 (2) a(i), b(ii), c(iv), d(iii)  
 (3) a(ii), b(i), c(iv), d(iii)  
 (4) a(ii), b(iii), c(iv), d(i)

86. Concentration of product formed after 2 min for given reaction is  
 $\text{A} \rightarrow \text{B}$

(Initial concentration of A is 2M and rate constant is  $2 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ )

- (1) 1.76 M  
 (2) 0.24 M  
 (3) 0.5 M  
 (4) 1 M

87. Propanal and propanone can be classified as

- (1) Position isomer  
 (2) Metamer  
 (3) Functional isomer  
 (4) Chain isomer

88. Most reactive compound towards nucleophilic addition reaction is

- (1) HCHO  
 (2)  $\text{CH}_3\text{CHO}$   
 (3)  $\text{C}_6\text{H}_5\text{CHO}$   
 (4)  $\text{CH}_3\text{COCH}_3$

89. Which among the following imparts apple green colour during flame test?

- (1)  $\text{Ca}^{2+}$   
 (2)  $\text{Cu}^{2+}$   
 (3)  $\text{Sr}^{2+}$   
 (4)  $\text{Ba}^{2+}$

90. Which among the following is Mohr's salt?

- (1)  $\text{Al}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$   
 (2)  $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$   
 (3)  $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$   
 (4)  $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

**BIOLOGY**

91. Match **Column-I** with **Column-II** and select the **correct** option.

|    | Column-I |       | Column-II     |
|----|----------|-------|---------------|
| a. | Mango    | (i)   | Muscidae      |
| b. | Wheat    | (ii)  | Polymoniales  |
| c. | Brinjal  | (iii) | Poales        |
| d. | Housefly | (iv)  | Dicotyledonae |

- (1) a(i), b(iv), c(ii), d(iii)  
 (2) a(iv), b(iii), c(ii), d(i)  
 (3) a(iv), b(iii), c(i), d(ii)  
 (4) a(iii), b(ii), c(iv), d(i)

92. Identify the **incorrect** statement w.r.t. Archaeobacteria.

- (1) They are responsible for the production of biogas from the dung of ruminant animals.  
 (2) They can survive in extreme salty areas and hot springs.  
 (3) They lack peptidoglycan in their cell wall.  
 (4) They differ from other bacteria in having different ribosomal subunits and cell wall structure.

93. Select the **odd** one out w.r.t. edible fungi.

- (1) Morels  
 (2) Truffles  
 (3) *Agaricus*  
 (4) Toadstools

94. Read the following statements and select the **correct** option.

**Statement A** : Flowers of *Cassia* and bean can be divided into two similar halves only in one particular vertical plane.  
**Statement B** : In *Asparagus*, the calyx and corolla are not distinct and are termed as perianth.

- (1) Only statement A is correct  
 (2) Only statement B is correct  
 (3) Both statements A and B are correct  
 (4) Both statements A and B are incorrect

95. Match **Column-I** with **Column-II** and select the **correct** option.

|    | Column-I      |       | Column-II       |
|----|---------------|-------|-----------------|
| a. | Monoadelphous | (i)   | <i>Citrus</i>   |
| b. | Epiphyllous   | (ii)  | Pea             |
| c. | Diadelphous   | (iii) | <i>Gloriosa</i> |
| d. | Polyadelphous | (iv)  | China rose      |

- (1) a(iv), b(iii), c(i), d(ii)  
 (2) a(iv), b(iii), c(ii), d(i)  
 (3) a(iii), b(iv), c(i), d(ii)  
 (4) a(ii), b(iv), c(i), d(iii)

96. Identify the **correctly** matched pair.

- (1) Stilt root – Banyan tree  
 (2) Prop root – *Rhizophora*  
 (3) Tap root – Wheat plant  
 (4) Storage root – Sweet potato

97. Which of the following statements is **not** true w.r.t. imperfect fungi?

- (1) Large number of them are decomposers of litter and help in mineral cycling  
 (2) The mycelium is aseptate and coenocytic  
 (3) They reproduce by asexual spores known as conidia  
 (4) *Trichoderma* is an example of these fungi

98. Read the following statements and state them as **True (T)** or **False (F)** and select the correct option.

A. Incomplete oxidation of glucose occurs in both prokaryotes and eukaryotes.  
 B. The Krebs's cycle starts with condensation of the acetyl CoA and citric acid.  
 C. In the partial oxidation of glucose, not much energy is released.

|     | A | B | C |
|-----|---|---|---|
| (1) | F | T | F |
| (2) | F | F | F |
| (3) | T | T | F |
| (4) | T | F | T |

- (1) (1)  
 (2) (2)  
 (3) (3)  
 (4) (4)

99. Select the **mismatched** pair.

|     |                                                           |   |                        |
|-----|-----------------------------------------------------------|---|------------------------|
| (1) | First stable product of photosynthesis in the maize       | – | Oxaloacetic acid       |
| (2) | Primary $\text{CO}_2$ acceptor in sugarcane               | – | Four carbon molecule   |
| (3) | First stable product of photosynthesis in the bell pepper | – | 3-phosphoglyceric acid |
| (4) | Primary $\text{CO}_2$ acceptor in tomato                  | – | Five carbon molecule   |

- (1) (1)  
(2) (2)  
(3) (3)  
(4) (4)

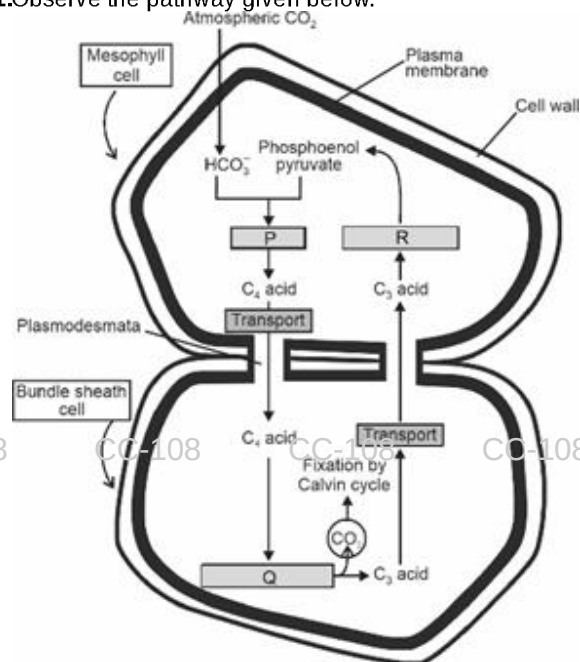
100. Read the following statements.

- A. Nutrient cycle and species composition of an ecosystem are its structural components.  
B. Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis.  
C. NPP depends on the variety of environmental factors and photosynthetic capacity of the plant.  
D. Only prokaryotic organisms are involved in breaking of dead remains of plants into the smaller particles by fragmentation process.  
E. The process of decomposition is largely an oxygen requiring process.

The **correct** ones are -

- (1) A and C  
(2) B, C and E  
(3) B and D  
(4) B, C, D and E

101. Observe the pathway given below.



Select the **correct** option to fill the blanks, which are indicated by P, Q and R.

|     | P             | Q               | R               |
|-----|---------------|-----------------|-----------------|
| (1) | Reduction     | Carboxylation   | Regeneration    |
| (2) | Fixation      | Regeneration    | Decarboxylation |
| (3) | Carboxylation | Regeneration    | Decarboxylation |
| (4) | Fixation      | Decarboxylation | Regeneration    |

- (1) (1)  
(2) (2)  
(3) (3)  
(4) (4)

102. In the process of glycolysis,

- (1) There is no release of water molecule  
(2) Net gain of ATP is four per glucose molecule  
(3) Invertase enzyme is involved  
(4) Substrate level phosphorylation reactions occur at two steps

103. What percentage of PAR is captured by the plants, which sustains the entire living world?

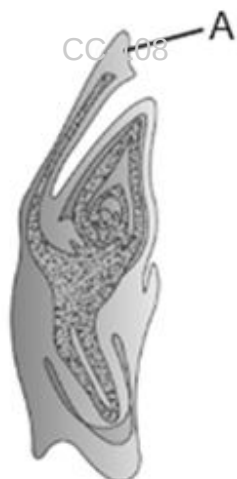
- (1) 1-5%  
(2) 2-10%  
(3) >50%  
(4) 99%

- 104.** Formation of which of the following is an example of dedifferentiation?
- Tracheary elements
  - Interfascicular cambium
  - Secondary phloem
  - Chlorenchyma
- 105.** Identify the **correct** statement from the following .
- Coriander and cotton show environmental heterophylly
  - N<sup>6</sup>-furfurylamino purine is an adenine derivative
  - Auxins are widely applied in tea plantations for branching
  - F. Skoog and his co-workers identified and crystallised kinetin
- 106.** Consider the given statements and choose the correct option.  
**Statement (A):** The per cent weight of an element that plays a very important role in blood clotting is high in Earth's crust than in human body.  
**Statement (B):** Ash analysis of a living tissue gives an idea of the types of inorganic elements present in the tissue.
- Both statements (A) and (B) are correct
  - Both statements (A) and (B) are incorrect
  - Only statement (A) is correct
  - Only statement (B) is correct.
- 107.** Choose the odd one w.r.t aromatic amino acids.
- Phenylalanine
  - Tryptophan
  - Valine
  - Tyrosine
- 108.** Palmitic acid differs from arachidonic acid in
- Being soluble in water
  - Lacking C=C double bonds
  - Possessing lower melting point
  - Having choline in its structure
- 109.** All of the following are polymeric substances as well as they belong to the category of secondary metabolites, **except**
- Rubber
  - Gums
  - Cellulose
  - Glycerol
- 110.** The type of epithelium that is present in the inner lining of oviducts possesses structures named 'X'. 'X' are also present in the epithelium lining the
- Bronchioles
  - Stomach
  - Small intestine
  - Proximal convoluted tubule
- 111.** Inspiration is possible if the pressure within the lungs is
- More than the atmospheric pressure
  - Equal to the atmospheric pressure
  - Twice the atmospheric pressure
  - Less than the atmospheric pressure
- 112.** Select the correct match w.r.t humans.
- Association areas – Control the urge for thirst
  - Thalamus – Major coordinating centre for sensory and motor signaling
  - Corpus callosum – Tract of nerve fibres that connect cerebellar hemispheres
  - Medulla oblongata – Possesses centres that control respiration and cardiovascular reflexes only
- 113.** The number of base pairs present in the palindromic sequence recognised by the first isolated restriction endonuclease is
- Seven
  - Three
  - Two
  - Six
- 114.** Which of the following is a direct method of gene transfer used for plants?
- Micro-injection
  - Biolistics
  - Ti-plasmid mediated transfer
  - Retrovirus mediated transfer
- 115.** The enzyme that facilitates the joining of sticky ends obtained after restriction digestion of a desired DNA is classified under which class of enzymes?
- Class IV
  - Class V
  - Class III
  - Class VI

116. All of the following statements are true **except**

- Cells of the anther wall layer that nourishes the
- (1) developing pollens, generally have more than one nucleus.
  - (2) Microspore tetrads are formed as a result of reductional division in the cells of sporogenous tissue.
  - (3) In less than 40 percent of angiosperms, generative cells undergo mitosis to produce two male gametes before the pollen grains are shed
  - (4) The distal end of the filament of stamen is attached to the thalamus or the petal of a flower.

117. Observe the figure given below.



Choose the **correct** option for 'A'.

- (1) Epiblast
- (2) Shoot apex
- (3) Scutellum
- (4) Coleoptile

118. Identify the **correct** statement.

- (1) Persistent, residual nucellus called perisperm is not found in beet
- (2) Flowers of *Salvia* have large feathery stigma to easily trap pollen grains.
- (3) *Vallisneria* prevents autogamy as well as geitonogamy
- (4) Endosperm gets partially consumed by the embryo and persists in the mature seed of groundnut

119. Which of the following statements is **incorrect** w.r.t. Mendel's experiments on pea plant?

- (1) In a true breeding tall or dwarf pea variety, the allelic pair of genes are homozygous.
- (2) Phenotype of  $F_1$  heterozygote differed from the parent with the dominant genotype in a monohybrid cross.
- (3) Recessive parental trait was expressed without any blending in the  $F_2$  generation.
- (4)  $F_1$  progeny produced four different types of gametes in a dihybrid cross.

120. Select the **correct** statement w.r.t. Thalassemia.

- (a) It is an autosomal recessive blood disease.
- (b)  $\beta$ -thalassemia is controlled by two closely linked genes on chromosome 16 of each parent
- (c) It is controlled by a single pair of allele,  $Hb^A$  and  $Hb^S$ .
- (d) It is quantitative problem of synthesising very few globin molecules.

In the light of the above statements, select the **correct** option from below.

- (1) Only (b) and (d) are correct
- (2) (a), (b) and (d) are correct
- (3) Only (a) and (d) are correct
- (4) (b) and (c) are correct

121. Match the List-I with List-II.

|    | List-I                       |       | List-II               |
|----|------------------------------|-------|-----------------------|
| a. | Skin colour in humans        | (i)   | Grasshopper           |
| b. | XO type of sex determination | (ii)  | Polygenic inheritance |
| c. | Female heterogamety          | (iii) | Pleiotropy            |
| d. | Phenylketonuria              | (iv)  | Birds                 |

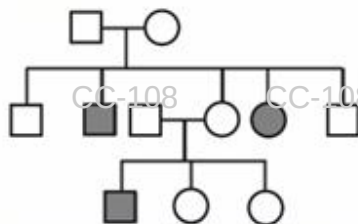
Select the **correct** option from below.

- (1) a(ii), b(i), c(iv), d(iii)
- (2) a(iii), b(iv), c(i), d(ii)
- (3) a(ii), b(i), c(iii), d(iv)
- (4) a(ii), b(iv), c(i), d(iii)

122. A male having AB blood group marries a female having O blood group. Which of the following options depicts the possible blood groups in their children?

- (1) A, B, AB and O
- (2) A and B
- (3) AB only
- (4) O only

123. Observe the pedigree chart given below.



The above pedigree chart represents a/an (A) trait, an example of which is (B).

Choose the **correct** option for (A) and (B) respectively.

- (1) Autosomal dominant trait, Myotonic dystrophy
- (2) Autosomal recessive trait, Sickle-cell anaemia
- (3) Sex-linked recessive, Colour blindness
- (4) Sex-linked recessive, Phenylketonuria

**124.** The meristems that are intercalated in between the permanent tissues is called

- (1) Cork cambium
- (2) Intercalary meristem
- (3) Intrafascicular vascular cambium
- (4) Shoot apical meristem

**125.** Read the given assertion (A) and reason (R) and the choose the **correct** option.

**Assertion (A) :** In stems of grasses, large sized vascular bundles are arranged in a ring-like manner.

**Reason (R) :** The large sized vascular bundles in monocot stem are densely arranged towards the periphery as compared to smaller ones.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

**126.** \_\_\_\_\_ is not found in guanosine.

In the light of above statement, choose the option which **correctly** fills in the blank.

- (1) Nitrogenous base
- (2) Phosphate
- (3) Sugar
- (4) N-glycosidic linkage

**127.** Read the given statements (A-D) and choose the **correct** option.

(A) In the first phase of translation binding of mRNA to the small subunit of ribosome occurs.

(B) In *E. coli*, mRNA does not require any processing to become active.

(C) In living systems, there are some biochemical reactions that are catalysed by RNA catalysts and not by protein enzymes.

(D) RNA polymerase is only capable of catalysing the elongation process.

How many among the given statements is/are **correct**?

- (1) Four
- (2) One
- (3) Two
- (4) Three

**128.** Select the option w.r.t. **correct** features of AUG codon.

- (a) It is an initiation codon.
- (b) It codes for tyrosine and methionine.
- (c) It shows degeneracy.
- (d) It has dual functions.

- (1) (a) and (c) only
- (2) (b) and (d) only
- (3) (a) and (d) only
- (4) All (a), (b), (c) and (d)

**129.** Select the **correctly** matched pair.

|     |                         |   |                                                            |
|-----|-------------------------|---|------------------------------------------------------------|
| (1) | Bean-shaped guard cells | – | Outer walls (away from stomatal pore) are highly thickened |
| (2) | Pericycle               | – | Gives rise to lateral roots                                |
| (3) | Isobilateral leaf       | – | Collenchymatous hypodermis                                 |
| (4) | Vessels                 | – | Radial conduction of water                                 |

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

**130.** The distance between two consecutive base pairs in DNA is 0.34 nm, then the length of DNA for a human diploid cell is turned out to be approximately

- (1) 1.5 metres
- (2) 1.36 metres
- (3) 2.2 metres
- (4) 2.6 metres

**131.** From the given steps, select the one that is usually performed after transferring the recombinant DNA into the host cell.

- (1) Restriction digestion of the desired DNA fragment
- (2) Transformation of the host cell
- (3) Ligation of desired DNA within the cloning vector
- (4) Culturing of the host cells at large scale

**132.** Granular bodies composed of RER and ribosomes are found in

- (1) Axon and cyton
- (2) Dendrites and synaptic knob
- (3) Synaptic knob and cell body
- (4) Dendrites and cell body

**133.** If a circular DNA has four and two restriction sites for only 2 restriction enzymes, i.e., *Pst* I and *Pvu* I respectively, then after complete restriction digestion of the DNA with *Pvu* II and *Pst* I, how many DNA fragments will be obtained?

- (1) Six
- (2) Seven
- (3) Four
- (4) Five

134. Consider the following statements w.r.t. humans.

- (a) Patella and sternum are the parts of appendicular skeleton.
- (b) Scapula is situated in the dorsal part of thorax between the second and seventh rib.
- (c) Thoracic vertebrae participate in the formation of ribcage.

Choose the **correct** option.

- (1) (a), (b) and (c) are correct
- (2) Only (a) and (b) are correct
- (3) Only (b) and (c) are correct
- (4) Only (c) is correct

135. Match the columns w.r.t. standard ECG in a healthy individual.

| Column I         | Column II                         |
|------------------|-----------------------------------|
| a. P-wave        | (i) Repolarisation of ventricles  |
| b. QRS complex   | (ii) End of ventricular systole   |
| c. T-wave        | (iii) Depolarisation of atria     |
| d. End of T-wave | (iv) Depolarisation of ventricles |

Select the **correct** option.

- (1) a-(iii), b-(ii), c-(iv), d-(i)
- (2) a-(iii), b-(i), c-(ii), d-(iv)
- (3) a-(iii), b-(iv), c-(i), d-(ii)
- (4) a-(i), b-(ii), c-(iii), d-(iv)

136. All of the following are components of the myofilament which are pulled towards the centre of 'A' band during muscle contraction, **except**

- (1) Troponin
- (2) Tropomyosin
- (3) 'F' actin
- (4) Light meromyosin

137. Which of the following animal is herbivore and characterised by presence of large bony plates on its back?

- (1) *Brachiosaurus*
- (2) *Stegosaurus*
- (3) *Pteranodon*
- (4) *Triceratops*

138. In humans, strength of ventricular contraction, cardiac output and heart rate can be increased by all of the following, **except**

- (1) Neural signals through sympathetic nerves fibres
- (2) Adrenal medullary hormones
- (3) Strenuous workout in gym
- (4) Neural signals through parasympathetic nerve fibres

139. Arrange the following in increasing order of their cranial capacities.

- (a) *Homo habilis*
- (b) Neanderthal man
- (c) *Homo erectus*

Select the **correct** option.

- (1)  $a < c < b$
- (2)  $a < b < c$
- (3)  $c < b < a$
- (4)  $c < a < b$

140. The seminiferous tubules of testis open into the vasa efferentia through

- (1) Epididymis
- (2) Ejaculatory duct
- (3) Vas deferens
- (4) Rete testis

141. Select the **incorrect** statement w.r.t. humans.

- (1) Zygote is the structural and functional link between developing embryo and maternal body.
- (2) Sperm formation continues even in old men.
- (3) The male sex accessory glands include paired bulbourethral glands and unpaired prostate gland.

The parts of female reproductive system along with a pair of mammary glands are integrated structurally and functionally to support the process of ovulation, fertilisation, pregnancy and child care.

- (4) The parts of female reproductive system along with a pair of mammary glands are integrated structurally and functionally to support the process of ovulation, fertilisation, pregnancy and child care.

142. Choose the odd one w.r.t. the structures which are cellular aggregations around nerve endings in frogs.

- (1) Sensory papillae
- (2) Taste buds
- (3) Nasal epithelium
- (4) Eyes

143. In humans, the epididymis leads to vas deferens that ascends to the abdomen and loops over the

- (1) Bulbourethral gland and urethra
- (2) Structure that receives ureters from the kidneys
- (3) Duct from seminal vesicles
- (4) Structure made of special tissue which helps in erection

- 144.** In male cockroach, genital chamber lies at A of abdomen that contains dorsal B, ventral C and D.

Choose the **correct** option.

|     | A             | B                 | C                 | D                 |
|-----|---------------|-------------------|-------------------|-------------------|
| (1) | Anterior end  | Anus              | Gonapophyses      | Male genital pore |
| (2) | Hind end      | Gonapophyses      | Anus              | Male genital pore |
| (3) | Posterior end | Male genital pore | Gonapophyses      | Anus              |
| (4) | Hind end      | Anus              | Male genital pore | Gonapophyses      |

- (1) (1)  
(2) (2)  
(3) (3)  
(4) (4)

- 145.** Read the following statements carefully w.r.t humans.

- (a) A functional mammary gland is the characteristic of all the mammals.  
(b) Only primary sex organs produce gametes in both males and females.  
(c) Spermatogenesis in males and oogenesis in females begins at puberty due to release of sex hormones.  
(d) Spermatogonia multiply by mitotic divisions and increase in numbers.

Choose the option with **correct** statements only.

- (1) (a) and (b)  
(2) (b) and (c)  
(3) (c) and (d)  
(4) (b) and (d)

- 146.** Select the option which is **not true** for nicotine.

- (1) It is not an alkaloid  
(2) Stimulates adrenal medulla to release catecholamines  
(3) It is one of chemicals present in tobacco  
(4) Involved in raising blood pressure and increasing heart rate

- 147.** Which of the following is **true** for secretions of endocrine glands?

- (1) Are nutritious chemicals  
(2) Transported to distantly located target organs  
(3) Secreted by organised endocrine glands only  
(4) Produced in large amounts

- 148.** A boy on exposure to pollen grains started continuously sneezing along with watery eyes and running nose. Which of the following antibodies is predominantly produced in his body during this condition?

- (1) IgA  
(2) IgM  
(3) IgE  
(4) IgG

- 149.** A person shows the given symptoms

- (a) Increased BMR  
(b) Protruding eyeballs  
(c) Weight loss  
(d) Abnormal swelling on either side of trachea.

Select the disease with which the person might be suffering from.

- (1) Graves' disease  
(2) Addison's disease  
(3) Gigantism  
(4) Diabetes mellitus

- 150.** Which of the following options is not correct w.r.t. carcinogens that transform normal cells into neoplastic cells?

- (1) Activation of *c-onc* of normal cells  
(2) Ionising radiations like X-rays  
(3) Non-ionising radiations like gamma rays  
(4) Oncogenic viruses possessing viral oncogenes

- 151.** Complete the analogy by selecting the **correct** option.

Infectious disease : Polio :: Non-infectious disease :

- (1) AIDS  
(2) Tetanus  
(3) Diphtheria  
(4) Cancer

- 152.** A vertebrate named 'X' possesses cartilaginous vertebral column and cranium. 'X' lacks scales but possesses unpaired fins. Identify 'X' by selecting the correct option.

- (1) *Scoliodon*  
(2) *Betta*  
(3) *Myxine*  
(4) *Pterophyllum*

- 153.** Darwin's finches exhibit all of the following evolutionary phenomena, **except**

- (1) Divergent evolution  
(2) Adaptive radiation  
(3) Convergent evolution  
(4) Founder's effect

154. Consider the statements given below.

- Evolution is a directed process in the sense of determinism.
- Evolution is a stochastic process based on chance events in nature and chance mutation in the organisms.
- Bandicoot shows convergent evolution with wombat.
- Sweet potato and potato are examples of homology.

Which of the above statements are not true?

- (a) and (b) only
- (a), (c) and (d) only
- (c) and (d) only
- (b), (c) and (d) only

155. Tubular secretion is one of the processes involved in urine formation. Tubular secretion of which of the following substances do(es) not take place by PCT of nephron?

- $H^+$
- Ammonia
- $HCO_3^-$

Select the option with **correct** set.

- (a) and (b)
- (b) and (c)
- Only (c)
- (a), (b) and (c)

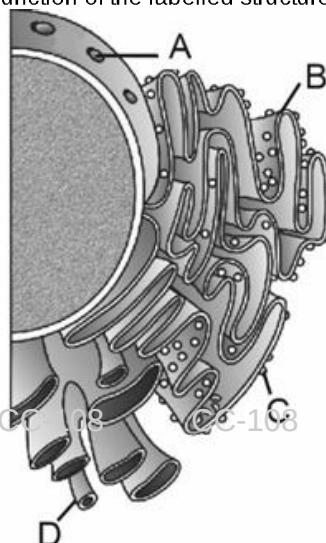
156. In eukaryotes, the regulation of gene expression could be exerted at all of the following levels, **except**

- The level at which formation of primary transcript occurs and regulation of splicing
- Translational level
- Transport of mRNA from nucleus to cytoplasm
- Reverse transcriptional level

157. Which of the following vacuoles provides buoyancy in prokaryotes?

- Sap vacuole
- Contractile vacuole
- Gas vacuole
- Food vacuole

158. Observe the figure given below and identify the correct feature/function of the labelled structures (A – D).



- 'A' helps in oxidative phosphorylation
- 'B' helps in steroidal hormone synthesis
- 'C' is composed of ribonucleic acid and proteins and are surrounded by a membrane
- 'D' helps in detoxification of drugs

159. Match List I with List II

|    | List I    |       | List II                                              |
|----|-----------|-------|------------------------------------------------------|
| a. | Nucleolus | (i)   | Infolding of mitochondrial inner membrane            |
| b. | Cristae   | (ii)  | Site for ribosomal RNA synthesis                     |
| c. | Cisternae | (iii) | Part of chromosome beyond secondary constriction     |
| d. | Satellite | (iv)  | Flattened sac-like structures stacked on one another |

Choose the **correct** answer from the following options.

- a(iii), b(ii), c(i), d(iv)
- a(ii), b(i), c(iv), d(iii)
- a(ii), b(i), c(iii), d(iv)
- a(iii), b(i), c(ii), d(iv)

160. During the "Pachytene" stage of meiosis

- Chromosomes start pairing and form bivalents
- There is formation of synaptonemal complex
- Crossing over between sister chromatids of homologous chromosomes occurs
- Four chromatids of each bivalent chromosomes become distinct and clearly appear as a tetrad.

**161.** Read the following statements of assertion (A) and Reason (R) and select the **correct** answer from the following options.

**Assertion (A) :** G<sub>0</sub> phase is called the quiescent stage of the cell cycle.

**Reason (R) :** Cells in this phase are metabolically active but no longer proliferate.

Select the **correct** answer from the following options.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

**162.** Astral rays arise from

- (1) Chromatids
- (2) Centromere
- (3) Kinetochore
- (4) Centrosome

**163.** Read the following statements (A to D) and select the option with all **correct** statements.

(A) The sex organs in first embryophytes are antheridium and archegonium.

(B) Gemmae are green, multicellular, sexual buds, which develop in gemma cups.

(C) Sporophyte in *Funaria* is more elaborate than that in *Riccia*.

(D) The predominant stage of the life cycle of a moss is the sporophyte which consists of two stages, protonema and leafy stage.

- (1) (A) and (D)
- (2) (A) and (C)
- (3) (B) and (D)
- (4) (B) and (C)

**164.** Select the **mismatched** pair.

|     |                  |                                                              |
|-----|------------------|--------------------------------------------------------------|
| (1) | <i>Pinus</i>     | Male cones and female cones are borne on same plant          |
| (2) | <i>Cycas</i>     | Male cones and megasporophylls are borne on different plants |
| (3) | <i>Salvinia</i>  | All the spores are of similar kinds                          |
| (4) | <i>Equisetum</i> | Presence of distinct compact structures called strobili      |

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

**165.** Once the BOD of sewage or waste water is reduced significantly, the effluent

- (1) Is pumped into large aeration tanks, where microbes consume the organic matter in the effluent
- (2) Is directly passed into anaerobic sludge digesters for biogas production
- (3) Is passed into a settling tank where the bacterial 'flocs' are allowed to sediment

After sedimentation is not clean enough to be released into natural water bodies due to its high polluting potential

**166.** Read the following statements and select the **correct** option.

**Statement A :** All birds and mammals are capable of maintaining homeostasis by physiological or behavioural means.

**Statement B :** Most snails and fish go into hibernation to avoid summer related problems.

- (1) Both statements are correct
- (2) Both statements are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

**167.** 'When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate'. This exemplifies

- (1) Habitat loss and fragmentation
- (2) Over-exploitation
- (3) Alien species invasion
- (4) Co-extinction

**168.** In each of the following sets, a conservation approach and an example of the method of conservation is given

- (a) *In-situ* conservation – Zoological park
- (b) *Ex-situ* conservation – Botanical garden
- (c) *In-situ* conservation – Sacred groves
- (d) *Ex-situ* conservation – Wildlife sanctuaries

Select the option with **correct** match of approach and method.

- (1) (a) and (b)
- (2) (a) and (c)
- (3) (a) and (d)
- (4) (b) and (c)

**169.** Citric acid is produced commercially from

- (1) *Acetobacter aceti*
- (2) *Clostridium butylicum*
- (3) *Aspergillus niger*
- (4) *Saccharomyces cerevisiae*

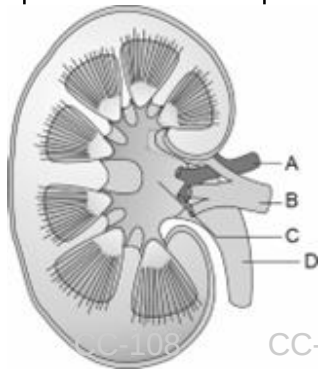
**170.** Which type of interaction is seen between sea anemone and clown fish?

- (1) Amensalism
- (2) Commensalism
- (3) Mutualism
- (4) Parasitism

**171.** Choose the odd one w.r.t. features that form the basis of differentiation between skeletal muscle fibres and muscle fibres present in urinary bladder.

- (1) Shape of muscle fibres
- (2) Position of nuclei
- (3) Presence of striations
- (4) Absence of branching

**172.** Observe the figure given below and select the correct option w.r.t. the labelled parts.



- (1) 'B' enters in kidney through hilum and carries oxygenated blood
- (2) 'C' is broad funnel shaped structure that forms renal columns in cortex region
- (3) 'A' exits from kidney through hilum and carries deoxygenated blood
- (4) 'D' carries urine which may be four times concentrated than the initial filtrate formed.

**173.** The animals included in the largest phylum of the kingdom Animalia have

- (1) Only two germ layers
- (2) Malpighian tubules as their excretory structures in all members
- (3) A segmented body
- (4) Closed circulatory system

**174.** Choose the mismatch.

- (1) Areolas connective tissue – Support framework for epithelium
- (2) Adipose tissue – Cells are specialised to store fats
- (3) Bone – Hard and pliable ground substance
- (4) Cartilage – Present in outer ear joints and between adjacent bones of the vertebral column

**175.** Match the column I with column II and select the correct option.

| Column I         |       | Column II                                    |
|------------------|-------|----------------------------------------------|
| a. Chiton        | (i)   | Body cavity not completely lined by mesoderm |
| b. Filarial worm | (ii)  | Organ-system level organization              |
| c. Locust        | (iii) | Radial symmetry                              |
| d. Squid         | (iv)  | Terrestrial                                  |
|                  | (v)   | Unsegmented body                             |

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(v), b(i), c(iv), d(ii)
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(ii), b(iv), c(ii), d(i)

**176.** Given below are two statements, one is the Assertion (A) and the other is Reason (R).

**Assertion (A):** Conversion of pro-insulin into mature insulin occurs in pancreatic cells where C-peptide is removed from a single polypeptide chain.

**Reason (R) :** The polypeptide chains in mature human insulin are linked together by phosphodiester bonds. In the light of above statements, select the most appropriate option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are false
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

**177.** Match column I and column II w.r.t transport of gases in humans.

| Column I                                             |       | Column II |
|------------------------------------------------------|-------|-----------|
| a. O <sub>2</sub> transported by RBCs                | (i)   | 7%        |
| b. CO <sub>2</sub> as Carbamino-haemoglobin          | (ii)  | 70%       |
| c. CO <sub>2</sub> as Bicarbonate ions               | (iii) | 20-25%    |
| d. CO <sub>2</sub> in dissolved state through plasma | (iv)  | 97%       |

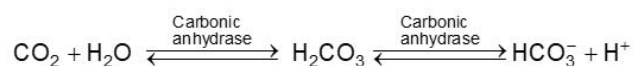
Select the **correct** option.

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(iv), b(ii), c(i), d(iii)
- (4) a(ii), b(iii), c(iv), d(i)

**178.** Choose the **incorrect** statement.

- (1) During the period of lactational amenorrhea, ovulation is inhibited.
- (2) Nirodh prevents users from contracting STIs.
- (3) Administration of progestogens within 72 hours of coitus have been found to be effective as emergency contraceptive.
- (4) Saheli is a steroidal oral contraceptive pill.

179. Which of the following is correct in context of following reaction?



- (1) At the alveolar site, formation of  $\text{HCO}_3^-$  is favoured
- (2) At the tissue site, formation of  $\text{CO}_2$  is favoured within RBCs using carbonic anhydrase
- (3) Carbonic anhydrase is present maximally in RBCs
- (4) Carbonic anhydrase is completely absent in blood plasma

180. Transmission and infection of HIV can be prevented by all of the following, **except**

- (1) Avoiding sex with multiple partners
- (2) Usage of sterile needles
- (3) By using condoms during coitus
- (4) By blood transfusion from infected individuals